Harshit Singh

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Experience

L&T Finance Mumbai, India
AI Platform Engineer *July 2025 – Present*

- **Gemma 3 4B Post-Training:** Leading post-training of Gemma 3 4B **SLM** for NBFC usecases; implementing fine-tuning methods (**SFT, RLHF, GRPO**). Conducting evaluation using **DCLM benchmarks** & domain-specific metrics; optimizing inference pipelines with **vLLM & llama.cpp**. Building **production-grade** understanding of **transformers** covering tokenization, pretraining, fine-tuning, evaluation & inferencing methods
- **TW Nostradamus Agentic AI Platform:** Built an agentic AI **workflow** (LangGraph) enabling conversational access to loan's credit risk & collection intelligence. Cut report analysis time by ∼**70%**. Integrated **GCP BigQuery** tables via **MCP server** to workflow delivering default risk, delinquency trends, and branch-wise KPIs
- **SMS & WA Monitoring:** Automated **end-to-end CRON** monitoring pipeline tracking hourly **SMS & WhatsApp** vendor performance; reduced manual efforts by **100%** and prevented collection drops by **10%**
- **Business-as-Usuals (BAUs):** Managed vendor lifecycle operations for 8+ concurrent vendors (Rs.8L+ monthly payouts); optimized cost-savings matrix achieving **Rs.0.53 Cr** monthly savings

LegalTrustNetwork Delhi, India
Software Development Intern *May 2024 – July 2024*

- **API Development & Product Features:** Conceived and wrote scalable **APIs** using **gRPC**; leveraged **gRPC-gateway** proxy for RESTful communication. Integrated new features & services on the Next.js web app.

Projects

Handwritten Equations to LaTeX - Seq2Seq Model | *PyTorch, CNN, LSTM, Seq2Seq*

- Developed an **image-to-LaTeX** neural network using **CNN-based encoder** for feature extraction & **RNN-based decoder with LSTM** for sequential LaTeX code generation
- Used **cross entropy loss** function & **teacher learning** for training the decoder, analysed **BLEU score**

AI QA Bot utilizing RAG | *LangChain, Deepseek-R1, OpenAI Embeddings, Chroma, FAISS*

- Built a **RAG** app using **OpenAI Embeddings** & **Deepseek-R1** on **LangChain CoT**, enabling answer retrieval
- Integrated **Chroma DB** to efficiently store and **FAISS** for retrieving embeddings, optimizing **semantic** searches

DriveSmart - AI-Powered Training for Safer Roads | *Scikit-Learn, KMeans, GMM, DBSCAN, PCA*

- Featured **time-series data** using **lag, differencing & rolling**; applied scaling, **PCA** dimensionality reduction
- Clustered driver's unsupervised data with **KMeans, GMM & DBSCAN**; optimized via **elbow & grid search**
- Assessed results via **silhouette/D-B scores** and derived behavioral profiles (risky, drowsy, etc.) from clusters

Rendezvous X – E-Ticketing Platform | *Django, MySQL, Docker, GitHub Actions, AWS*

- Led a 2-tier technical team of 5 for revamping the website and introducing e-commerce for merchandise sales
- Developed a scalable ticketing system for pronites using **Django & MySQL**, managed over **8k ticket bookings**
- Utilized **Docker**, **GitHub Actions** & **Elastic Beanstalk** to deploy webapp on **AWS**, achieved **99.9% uptime**

Technical Skills

Deep Learning & LLMs: PyTorch, ANN, RNN, CNN, NLP, LSTM, Transformers, HuggingFace, OpenAI API, LangChain/LangGraph, RAG, Vector Databases, MCP, PEFT, LoRA, RLHF, DPO, GRPO, vLLM, llama.cpp
Data & Cloud : GCP (BigQuery, Vertex AI, Cloud Run), AWS (EC2, ECS, Lambda, S3, RDS), Docker, FastAPI, n8n
Machine Learning: Scikit-Learn, XGBoost, LightGBM, NumPy, Pandas, Matplotlib
Languages: Python, Javascript/TypeScript, SQL, Bash

Education

Indian Institute of Technology Delhi New Delhi, India
Bachelor of Technology in Textile & Fibre Engineering *Aug. 2021 – May 2025*

Deep Learning Algorithms Stanford University
Supervised Machine Learning Stanford University